



LQES Introduction

i18n-L10n QA Event Support Service

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i18n QE

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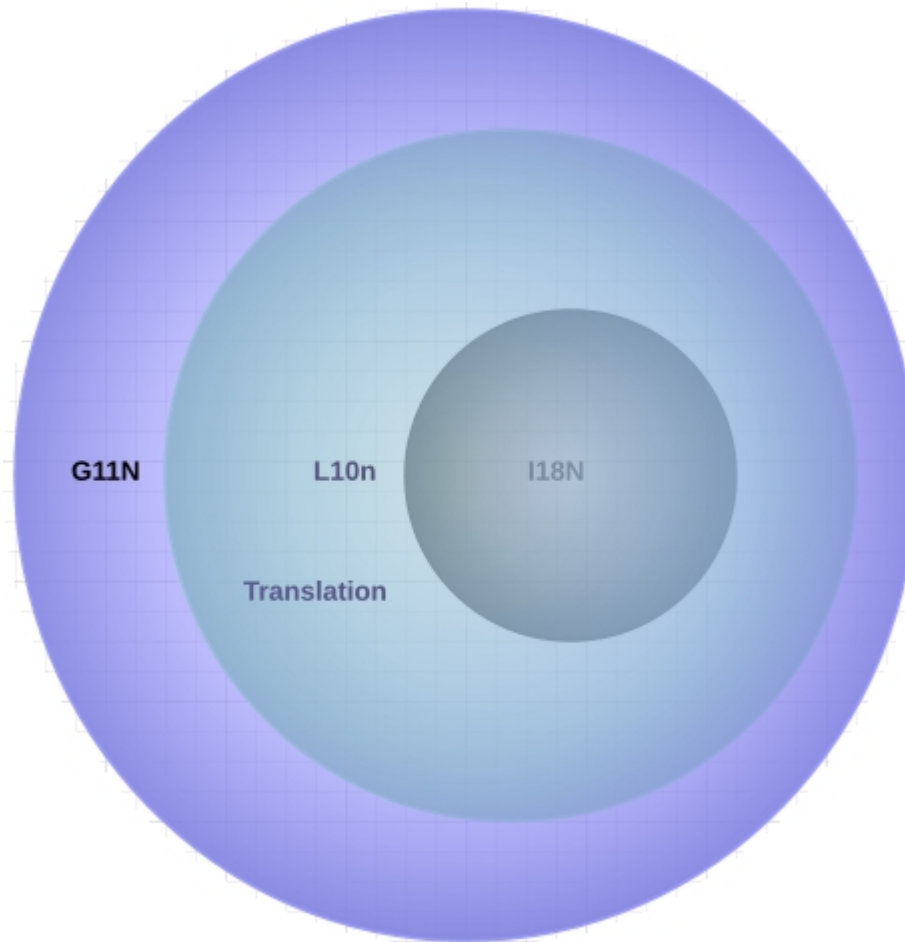
Sept 16th 2013

Agenda

- GILT
- What is i18n testing ?
- What is L10n testing ?
- Why need LQES ?
- How to execute TCMS test runs ?
- Reporting a New Bug via Red Hat Bugzilla
- Q&A

GILT

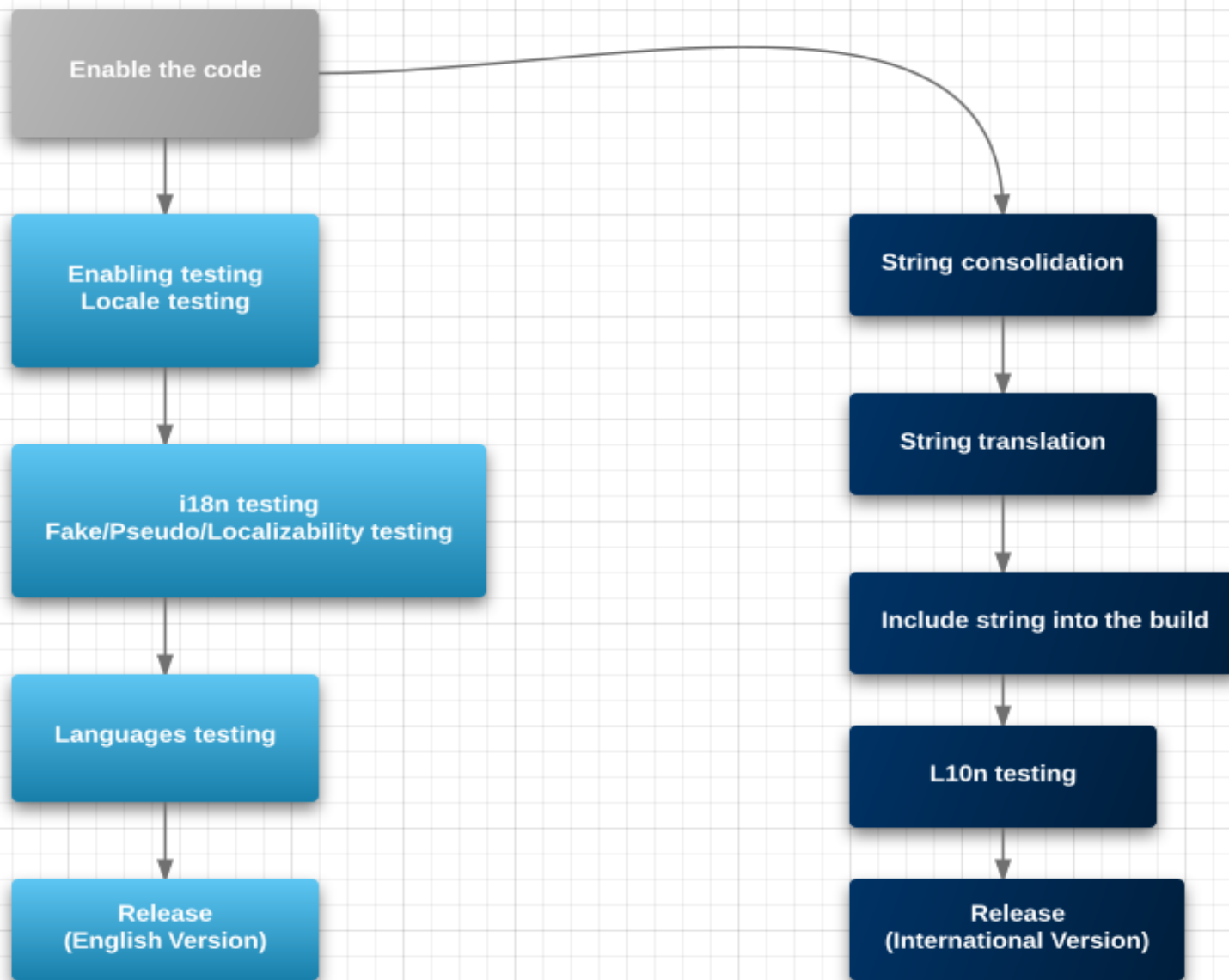
GILT: Globalization, Internationalization, Localization, Translation



What is i18n testing ?

- Internationalization (i18n) is the process of planning/designing/implementing a software application/websites/databases/products and services so that the underlying source code is locale-neutral and it can easily be adapted to various languages and regions/cultures/locales without engineering changes, the result is one generic code that can be used across locales. It makes sure that the code can handle all international support without breaking functionality that would cause either data loss or display problems.
- i18n is a required step before actual product localization may begin.
- i18n testing is done to ensure that the software does not assume any specific language or conventions associated with a specific language and to ensure the software will work globally regardless of locale. It checks proper functionality of the product with any of the culture/locale settings using every type of international input possible.

Major activities in i18n testing



i18n testing focus on:

Localizability testing:

'a, use à or å c, use â or ç n, use ñ or ñ, aa+++some strings+++bb'

- UI message extract (hard-coding, concatenation, stripping, text fragmentation, ambiguity, fonts)
- UI layout (Alignment, truncation, overlap, dialog resizing, space limitations, image layering and size information)
- Over translation(ex. consistent, excessive technical jargon)
- Mirroring testing (ex. Right-to-left (RTL) text, mirrors windows/text alignment.)
- It does not test for localization quality.

Input/output & functionality testing:

- Using Non-ASCII (non-English) characters (ex. East Asian languages, German, Complex Script characters: Arabic, Hebrew, Thai, and etc.),
- Install/Uninstall/Upgrade product on English/Localized OS,
- Character sets/Encoding Support (Unicode&Native, ex. single-byte character codes/set (SBCS) such as English, double-byte/multiple-byte character codes/set (DBCS/MBCS) such as Japanese Kanji, European characters such as German, bidirectional character set testing (BIDI) such as Hebrew or Arabic.)
- Multi-byte Data input/output
- Functionality loss

Locale testing:

- Formatting of Date/time, numbers, currency, calendars differences, measurement units, graphics, sorting, conversion, address, phone number, name, print, page size, Short-cut/hot key, bidi, white space, and etc
- GUI/message display (ex. Garbage characters)
- Searching with non-ASCII characters
- IME (ex. Ibus & local IME such as kkc, libpinyin, and etc)
- Time Zone
- Display issues (ex. Question marks (?): Unicode-to-ANSI conversion; Random High ANSI characters (e.g., ¼, †, ‰, ‡, ¶): ANSI code using the wrong code page.; boxes, vertical bars, or tildes (default glyphs) [□, |, ~]: Selected font cannot display some of the characters.)

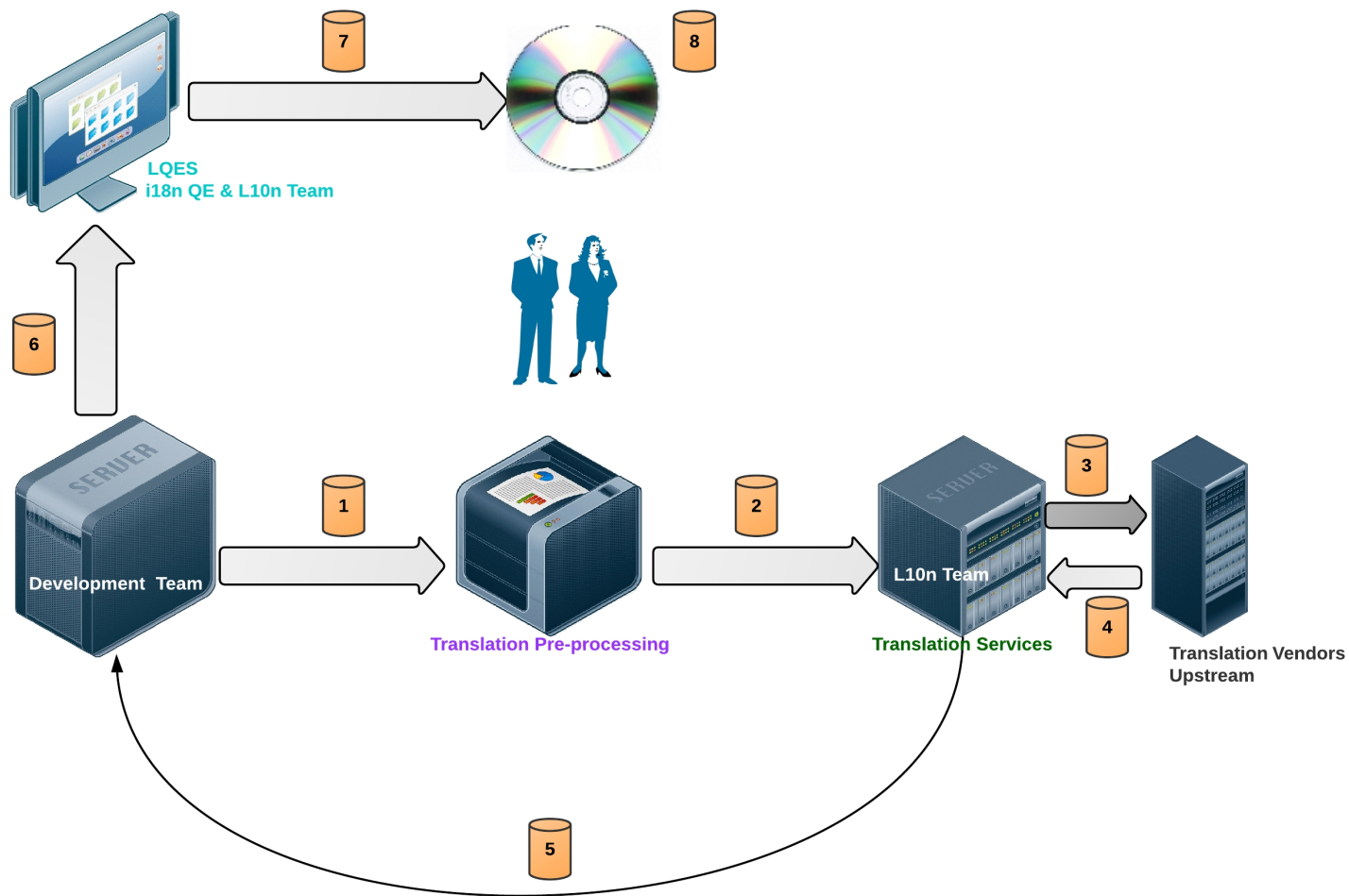
Languages testing

- Focuses on testing the English product with a global environment of products and services functioning in non-English.

What is L10n testing ?

- Localization (L10n) is the process of adapting software for a specific region or language by adding locale-specific components and translating text.
- L10n testing is based on the results of globalization testing, which verifies the functional support for that particular culture/locale.
- L10n testing checks the quality of a product's localization for a particular target culture/locale. Localization testing can be executed only on the localized version of a product.

Red Hat Translation Process Example



1. Dev. Team packaging the files (po files) and sends it to Translation pre-processing (English source strings).
2. Sending the English source strings to L10n team.
3. L10n team and then start the translation within Zanata/Transifex.
4. L10n team sync up translation with upstream.
5. Dev. Team include the localized strings to build from Zanata/Transifex .
6. Dev. Team build a multi-lingual version of the product.
7. I18n QE & L10n Team perform the LQES.
8. After LQES has successfully completed the product is ready for distribution.

L10n testing focus on:

Functionality testing

- › Basic functionality (locale-sensitive) tests on localized Env.
- › Hot key, shortcut key, mnemonic key accessibility
- › Check help via Help /F1 button

Locale/Data format/Encoding testing

- › Consistency of formats: Date/time, numbers, currency, calendars differences, measurement units, graphics, sorting, conversion, address, phone number, name, print, page size, bidi, white space, and etc.
- › DBCS/MBCS strings input/out, display
- › Font changes, encoding/codepage issues

UI/Cosmetic testing:

- Matching/Consistent with English version (printed documentation, online help, messages, interface resources, command-key sequences, etc.)
- Un translated text
- Hard code
- Text expansion/ Text that takes up two lines
- Characters appear as empty squares
- Un translated Tooltips
- Misalignment
- Truncation
- Overlap
- Layout/Appearance/Size
- String corruption/garble
- Punctuation
- Fields reversed or in the wrong order in right-to-left languages (such as Arabic, Urdu, Farsi, and Hebrew)

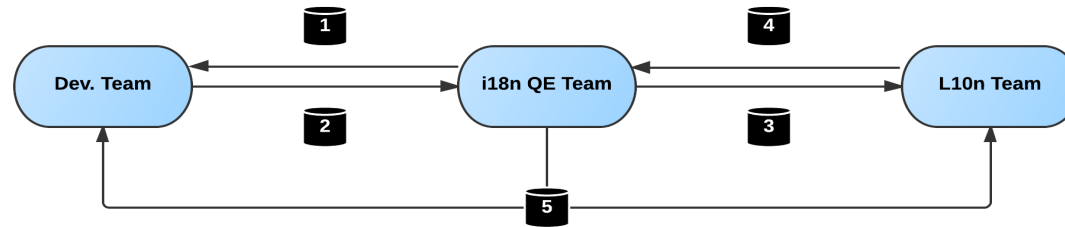
Translation/Linguistic testing

- All UI/Message strings are translated (manuals, online help, context help, etc.)
- Out-of-context translation errors
- linguistic accuracy and resource attributes
- Inadvertently translated strings
- Typographical, grammatical, and contextual errors (e.g. Is the tab order from right to left in right to left languages?)
- Punctuation
- No over translated.
- Terminology is used consistently in all documents and application UI

Why need LQES ?

- Make sure products i18n testing PASS by i18n QE team.
- Make sure products L10n testing PASS by native speaker L10n team.
- Make sure products Global - ready GA Release.

LQES Flowchart



1. I18n QE send the request to Dev. Team for localized build& testing Env.
2. Dev. Team give the localized build& testing Env to i18n QE team.
3. I18n QE prepare test cases/test runs and send the request to Dev./L10n team for reviewing the test cases; i18n testing by QE team; Create L10n QA Event Mojo page and send testing request to L10n team.
4. L10n team edit test runs and assign the testers, finishing the test runs.
5. I18n QE team summary/send out the test result report & QA event feedback; i18n QE & L10n team handling bug NEED_INFO and verify bug.

How to execute TCMS test runs ?

■ Mojo L10n QA Event Page

RHEVM 3.3 L10n QA Event (Sep. 17 - 24)

created by [Lijun Li](#) on Sep 15, 2013 6:38 AM, last modified by [Lijun Li](#) on Sep 15, 2013 6:38 AM

Test Env.

Test Runs

Admin Portal Test Runs

ja_JP <https://tcms.engineering.redhat.com/run/84329/>

■ Status & Comments to the test case in TCMS test run.

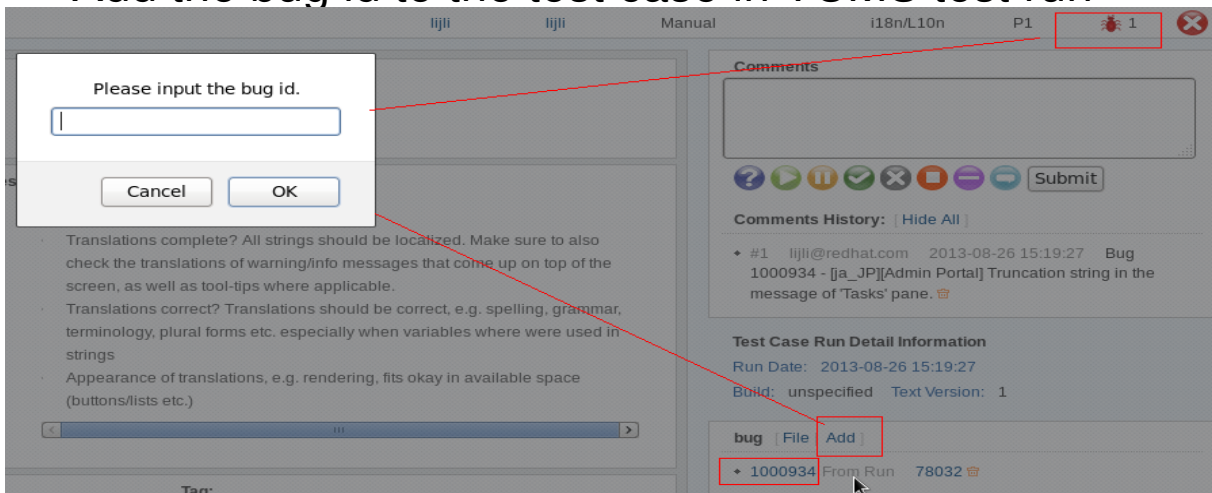
Comments



Comments History: **FAILED**

♦ #1 [lijli@redhat.com](#) 2013-08-26 15:19:27 Bug
1000934 - [ja_JP][Admin Portal] Truncation string in the
message of 'Tasks' pane. 🗑

■ Add the bug id to the test case in TCMS test run

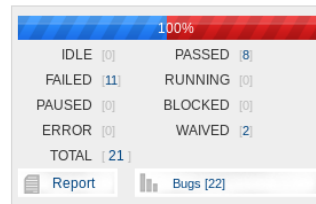


■ Click on top 'Set to Finished' button when finishing the TCMS test runs.

[Edit](#) [Clone](#) [Delete](#) [Export To CSV](#) [Export To XML](#)

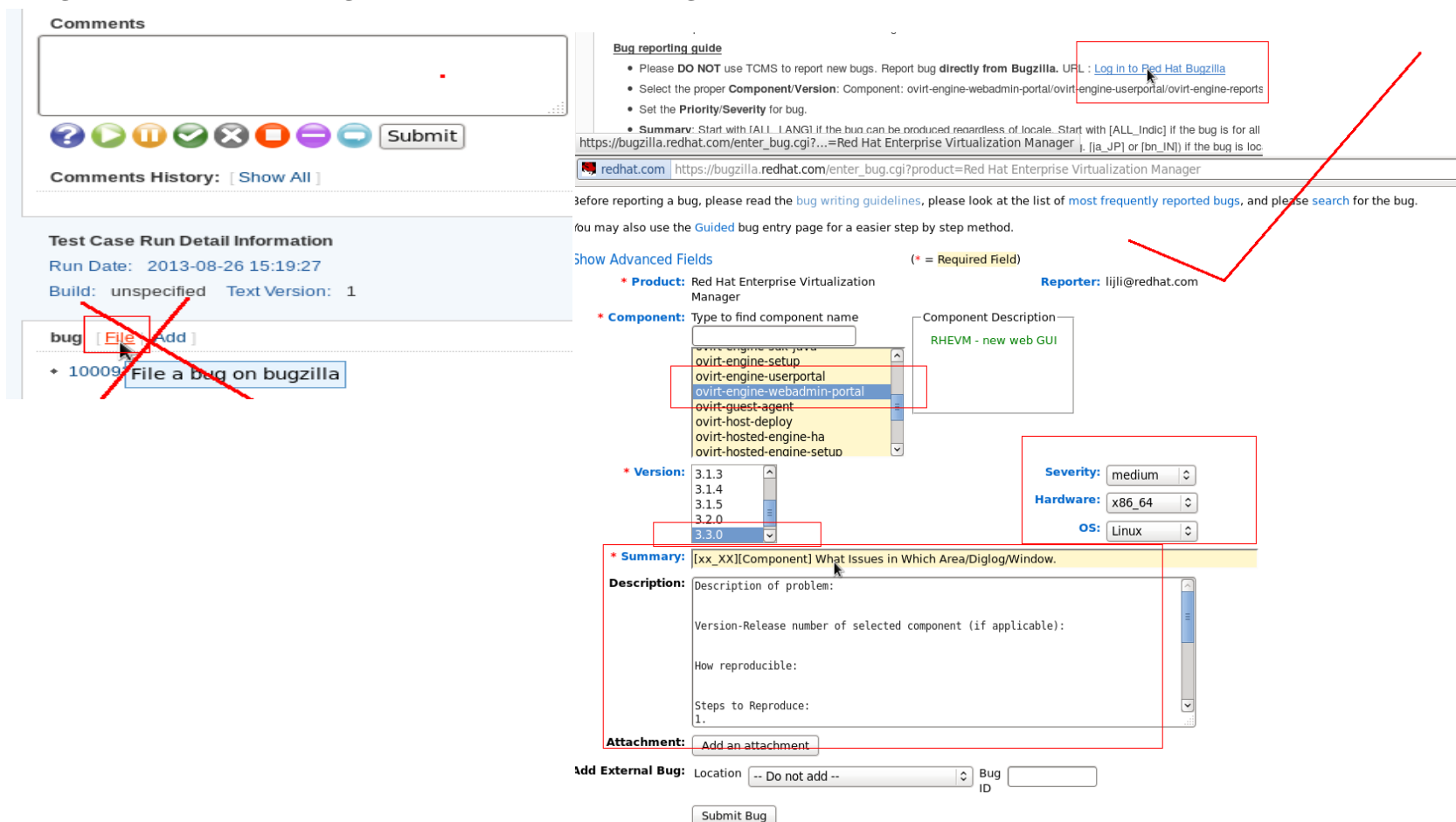
[ja_JP] [Admin Portal] Test run for i18n-L10n RHEVM 3.3 Test Plan on i18n

Test Plan : [\[10441\] i18n-L10n RHEVM 3.3 Test Plan](#)
Product Version : 3.3
Manager : lijli@redhat.com
Estimated Time : 0:00:00
Started at : 2013-08-22 11:49:47
Tags : [Add Tag](#)
Environment : [Add Property](#)
Note :
Product : Red Hat Enterprise Virtualization Manager
Build : unspecified
Errata :
Default Tester : lijli@redhat.com
Status : Running [Set to Finished](#)
Finished at :
CC : [Add CC](#)
Set Status Automatically : False



Reporting a New Bug via Red Hat Bugzilla

- Please DO NOT use TCMS to report new bugs. Report bug directly from Bugzilla. URL : Log in to Red Hat Bugzilla



The screenshot shows the Red Hat Bugzilla bug reporting interface. On the left, there is a 'Comments' section with a text area and a 'Submit' button. Below it is the 'Test Case Run Detail Information' section, showing 'Run Date: 2013-08-26 15:19:27', 'Build: unspecified', and 'Text Version: 1'. A red 'X' is drawn over the 'bug' link and the 'File a bug on bugzilla' button. On the right, there is a 'Bug reporting guide' section with instructions. A red box highlights the 'Log in to Red Hat Bugzilla' link. Below the guide, there is a 'Show Advanced Fields' section with various dropdown menus and text boxes. A red box highlights the 'Product' dropdown, which is set to 'Red Hat Enterprise Virtualization Manager'. Another red box highlights the 'Component' dropdown, which is set to 'ovirt-engine-webadmin-portal'. A third red box highlights the 'Version' dropdown, which is set to '3.3.0'. A fourth red box highlights the 'Severity' dropdown, which is set to 'medium'. A fifth red box highlights the 'Hardware' dropdown, which is set to 'x86_64'. A sixth red box highlights the 'OS' dropdown, which is set to 'Linux'. A red arrow points from the 'Log in to Red Hat Bugzilla' link to the 'Reporter' field, which is set to 'lijli@redhat.com'. At the bottom, there is an 'Add External Bug' section with a 'Location' dropdown and a 'Bug ID' text box. A 'Submit Bug' button is at the very bottom.

Comments

Comments History: [Show All]

Test Case Run Detail Information

Run Date: 2013-08-26 15:19:27

Build: unspecified Text Version: 1

bug [File] Add

10009 File a bug on bugzilla

Bug reporting guide

- Please DO NOT use TCMS to report new bugs. Report bug directly from Bugzilla. URL : [Log in to Red Hat Bugzilla](#)
- Select the proper Component/Version: Component: ovirt-engine-webadmin-portal/ovirt-engine-userportal/ovirt-engine-reports
- Set the Priority/Severity for bug.
- Summary: Start with [ALL_LANG] if the bug can be produced regardless of locale. Start with [ALL_Indic] if the bug is for all https://bugzilla.redhat.com/enter_bug.cgi?...=Red Hat Enterprise Virtualization Manager, [ja_JP] or [bn_IN] if the bug is loc

redhat.com https://bugzilla.redhat.com/enter_bug.cgi?product=Red Hat Enterprise Virtualization Manager

Before reporting a bug, please read the [bug writing guidelines](#), please look at the list of [most frequently reported bugs](#), and please [search](#) for the bug.

You may also use the [Guided](#) bug entry page for a easier step by step method.

Show Advanced Fields (* = Required Field)

- * Product: Red Hat Enterprise Virtualization Manager
- * Component: Type to find component name
ovirt-engine-setup
ovirt-engine-userportal
ovirt-engine-webadmin-portal
ovirt-guest-agent
ovirt-host-deploy
ovirt-hosted-engine-ha
ovirt-hosted-engine-setup
- * Version: 3.1.3
3.1.4
3.1.5
3.2.0
3.3.0
- Severity: medium
- Hardware: x86_64
- OS: Linux
- * Summary: [xx_XX][Component] What Issues in Which Area/Diglog/Window.
- Description: Description of problem:
Version-Release number of selected component (if applicable):
How reproducible:
Steps to Reproduce:
1.
- Attachment: Add an attachment

Add External Bug: Location -- Do not add -- Bug ID

Submit Bug

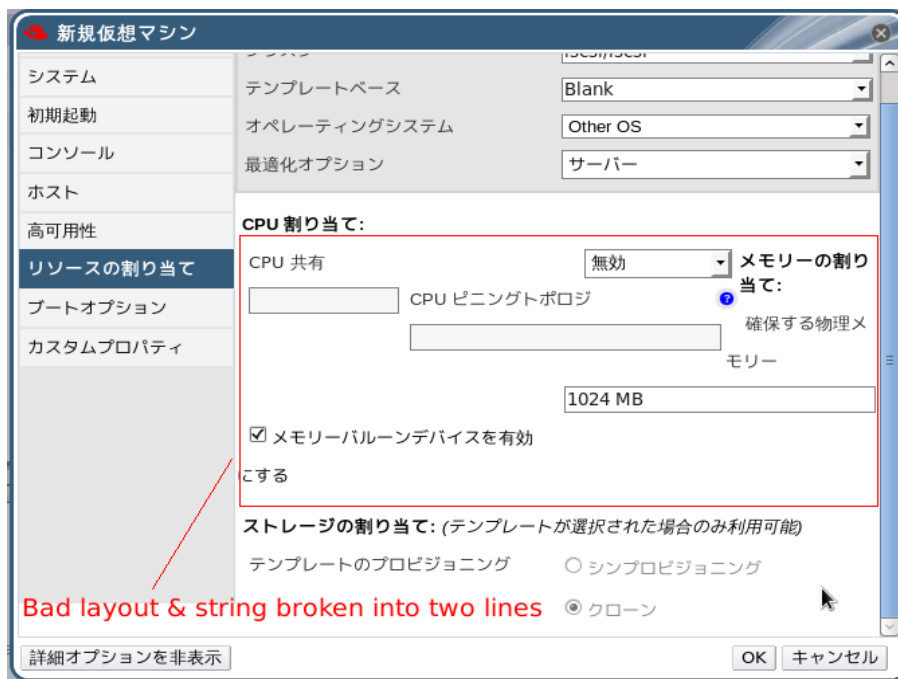
- Select the proper Component/Version: Component: (ex. ovirt-engine-webadmin-portal/ovirt-engine-userportal/ovirt-engine-reports Version: 3.3.0)
- Set the Priority/Severity for bug.
- Summary: Start with [ALL_LANG] if the bug can be produced regardless of locale, Start with [ALL_Indic] if the bug is for all Indic locales, Start with [CJK] if the bug is only affecting the CJK locales, otherwise, start with [LL_RR] (e.g. [ja_JP] or [bn_IN]) if the bug is locale specific.
Format: [xx_XX][Component]What issues in Which area/dialog/window.
- Write a clear Steps to Reproduce in Description.

Bug 1000934 - [ja_JP][fr_FR][Admin Portal] Truncation string in the message of 'Tasks' pane. (edit)**Status:** NEW (edit)**Aliases:** None (edit)**Product:** Red Hat Enterprise Virtualization Manager**Component:** ovirt-engine-webadmin-portal (Show other bugs)**Version:** 3.3.0**Hardware:** x86_64 Linux**Priority:** medium **Severity:** low**Target Milestone:** ---**Target Release:** 3.4.0**Assigned To:** Greg Sheremeta (edit) (take)**QA Contact:** Yuko Katabami (edit) (take)**Docs Contact:** (edit) (take)**URL:****Whiteboard:** ux**Keywords:** 18n, Improvement**Reported:** 2013-08-26 03:16 EDT by Lijun Li**Modified:** 2013-09-05 14:38 EDT (History)**CC List:** ☐ Add me to CC list
7 users**Add**
acathrow@redhat.com
bazulay@redhat.com
ecohen@redhat.com
iheim@redhat.com
qe-i18n-bugs@redhat.com☐ Remove selected CCs**See Also:** (add)**Fixed In Version:****Customer Escalation:** ---**Internal Whiteboard:****Devel Whiteboard:**

- Add 'qe-i18n-bugs@redhat.com' in the CC List for all reported bugs.
- Keywords: i18n if the bug is related to i18n. [i18n: focus on i18n issues like function, font, strings display (characters en-coding), garbage strings, input/output, sorting, GUI layout, truncation, date/time/currency/locales format and etc.]
- Keywords: Translation and add 'eng-l10n-bugs@redhat.com' in the CC list if the bug is related to L10n. [Translation: only focus on the string itself, including translation error, untranslated, inconsistent translation, and etc.]
- Bug type: Translations, unlocalized, truncation, overlapped, misalignment, garbage strings, font issues, punctuation symbol, date/time format, sorting, input/output for locale characters, and etc.
- For any confusion or help, feel free to contact smaitra@redhat.com
lijli@redhat.com (IRC channel: lijli, Satya @ #lqes)
- Don't accumulate several issues into a single BZ item - document one issue per BZ item
- Don't mark BZs as "urgent" severity unless it is justified to do so:
"urgent" severity usually means that everyone are stopping everything that they are doing and making sure that this BZ is taken care of, which isn't necessarily the case for each i18n-related bug.

Screen Capture/Screen Shot

- Add the Attachment (Screenshot): Please only take the screenshot on the issue dialog (relevant area) but not for all the desktop and highlight the issues with red rectangle (for verifying the bugs with green rectangle and saying fixed.) within the screen-shot. Better to provide the relevant context in English ('context' = name of main tab / sub-tab / dialog / button / etc.) and/or provide the parallel English screen-shot.



Q&A
Thank You